

Learning chemistry

Chemistry is concerned with matter, the stuff that every object in the Universe is made of. It is what many processes key to modern life are based on, from improving the quality of food in the supermarket to the obtaining of gas for a car. The study of chemistry became popular in Europe in the 1700s when scientists began to investigate substances, such as gases and liquids, and how they change. They carried out experiments to isolate and identify the elements (substances that cannot be broken down into simpler substances), and paved the way for future experiments and discoveries.



Knowledge of metals

This picture comes from *De re metallica* (On the Nature of Metals), an influential early book on chemistry published in 1556 in Germany. It described the extraction of metals by smelting (heating and melting).

Key events

1661

In his book *The Sceptical Chymist*, Anglo-Irish philosopher Robert Boyle argued that matter is made up of different "corpuscles" (atoms) that are constantly in motion.

1754

Joseph Black, a Scottish chemist, isolated the gas carbon dioxide, which he called "fixed air." He noted it was heavier than air, put out flames, and suffocated animals.

1766

English chemist Henry Cavendish discovered a colorless gas, which he called "inflammable air"—now known as hydrogen.

1789

Antoine Lavoisier published *Traité élémentaire de chimie* (Elementary Treatise of Chemistry), the first modern chemistry textbook.

Henry Cavendish

Alchemist's laboratory

Medieval alchemists heated and distilled substances in vials and retorts (glass vessels), and mixed and pounded them in mortars, all in the hope of turning lead into gold. They are often regarded as the early pioneers of chemistry.



An alchemist working in a laboratory



Crude oil is heated before reaching the column.



Chemistry experiment

In the early 1780s, French chemist Antoine Lavoisier (see p.107) designed the ice calorimeter for an experiment to show that respiration (breathing) is chemically the same process as combustion (burning)—they utilize oxygen and produce heat. The device measured the amount of heat a guinea pig produces when it breathes.